Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building SQL-backed operational systems, object-oriented applications, and testable deployment workflows for logistics-style, data-intensive environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Designed and optimized relational schemas across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+ records**, reducing dashboard latency by **71%** (**1.2s to 350ms**) through SQL tuning and indexed access patterns.
- Built data integration and deployment workflows with **FastAPI**, SQL Server, and cloud storage, processing **5K+ monthly jobs** with **99%+** reliability through validation, error handling, and repeatable release processes.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered cross-platform application software with reusable OOP modules and **20+** automated unit/integration suites using **Jenkins, PyTest, and Mockito**, supporting build, test, and delivery across Windows, Linux, and macOS.
- Optimized API and ETL flows with **SQL, AWS Glue, and S3**, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%** while documenting delivery and validation steps.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a **microservice-based operations platform** with **Azure SQL**, WebSockets, and REST APIs, supporting **8** business services for resource allocation, billing, permissions, notifications, and reporting across **2M+ records**.
- Automated testing and deployment with **Jenkins, Docker, Jest, Mocha, and Selenium**, improving successful build validation to **95%+** while maintaining structured technical workflows and release readiness.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Designed end-to-end backend workflow for a service-booking platform, mapped **12+ integration points**, and built **Flask** APIs for booking, pricing, coupon, and payment logic across **50K+ records**.
- Implemented route-like operational flows with real-time sync, webhooks, and analytics using **Firebase, Databricks, and Spark**, sustaining sub-**200ms** APIs and reducing manual refund handling by **70%**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Built backend workflows for a production platform handling **10K+ requests/day** with **Flask, MySQL, AWS RDS, and REST APIs**, supporting secure ingestion, batch execution, and stable deployment behavior.
- Maintained structured evaluation and tracking workflows for **150+ instructors**, reducing turnaround time by **60%** while improving reliability, documentation, and repeatability of operational tasks.

Skills

Languages: SQL, C++, Java, Python, JavaScript/TypeScript
Application Engineering: Object-Oriented Design, Technical Design, Coding, Testing, Deployment, Technical Documentation
Backend & Data: PostgreSQL, MySQL, SQL Server, Azure SQL, Flask, FastAPI, REST APIs, Triggers/Packages-style SQL Logic
Tools & Delivery: Jenkins, Docker, Git, CI/CD, AWS, Databricks, Spark, Cross-Platform Validation, Agile/Scrum

Projects

Chat2SQL GitHub | *SQL, PostgreSQL, Java, TypeScript*

- Built a structured SQL workspace supporting **10+** databases, enabling schema exploration, query generation, and relational workflow execution across Java and TypeScript modules with maintainable deployment workflows.

CPU-scheduling Simulator Live | *C++, React.js, Scheduling Algorithms*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** algorithms in a simulation tool, reducing debugging time by **20%** and improving average turnaround time by up to **25%** across **1K+** test scenarios.

Big-Brain Live | *PostgreSQL, REST APIs, Data Workflows*

- Built a data-heavy application over **50K+ records** with structured indexing, metadata persistence, and sub-second retrieval, demonstrating scalable design for operational analysis workflows.